

File Permissions & File Operations Challenge

- ◆ Task 1: Create Files
- ◆ Create empty file devops.txt using touch
- ◆ Create notes.txt with some content using cat or echo
- ◆ Create script.sh using vim with content: echo "Hello Devops"
- ◆ Verify ls -l to see permissions.

touch devops.txt: command is used to create the text file.

echo "This is my day 10 assignment notes."> notes.txt: command is used to create text file and passing input using the echo.

vim script.sh: command is used to create shell files.

```
ubuntu@ip-172-31-42-193:~$ touch devops.txt
ubuntu@ip-172-31-42-193:~$ echo "this is my day 10 assignment notes." > notes.txt
ubuntu@ip-172-31-42-193:~$ vim script.sh
ubuntu@ip-172-31-42-193:~$ █

ubuntu@ip-172-31-42-193:~$ ls -l devops.txt
-rw-rw-r-- 1 ubuntu ubuntu 0 Feb  9 09:51 devops.txt
ubuntu@ip-172-31-42-193:~$ ls -l notes.txt
-rw-rw-r-- 1 ubuntu ubuntu 36 Feb  9 09:52 notes.txt
ubuntu@ip-172-31-42-193:~$ ls -l script.sh
-rw-rw-r-- 1 ubuntu ubuntu 22 Feb  9 09:53 script.sh
ubuntu@ip-172-31-42-193:~$ █
```

- ◆ Task 2: Read Files
- ◆ Read notes.txt using cat
- ◆ View script.sh in vim read-only mode
- ◆ Display first 5 lines of /etc/passwd using head
- ◆ Display last 5 lines of /etc/passwd using tail

cat notes.txt: command is used to read the files.

```
ubuntu@ip-172-31-42-193:~$ cat notes.txt
this is my day 10 assignment notes.
```

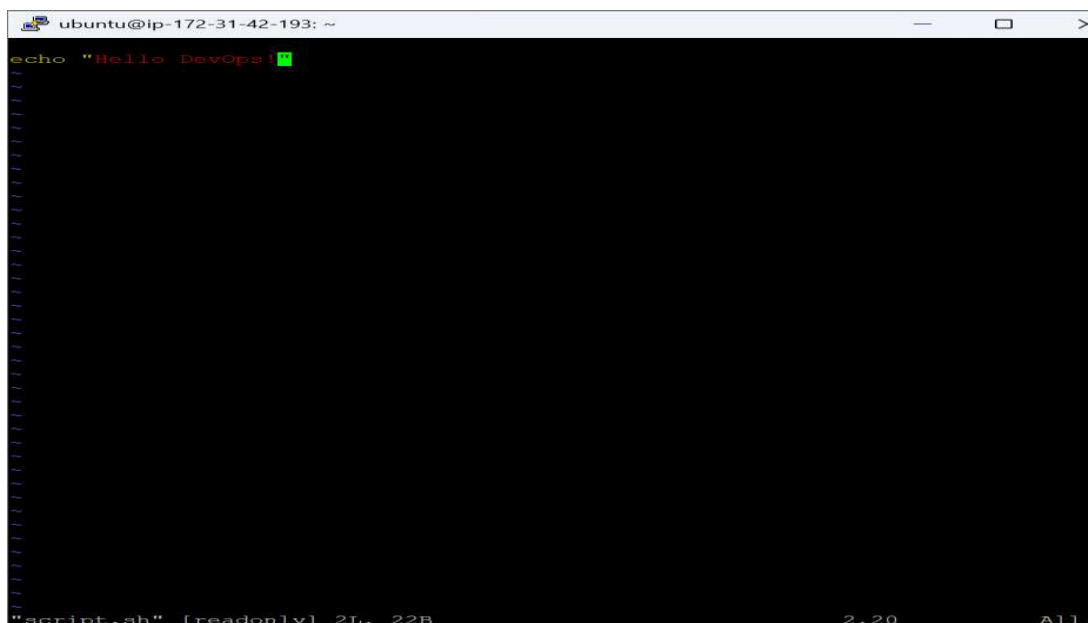
`head -n 5 /etc/passwd`: this file is world-readable and lists all system and regular user accounts, including system services that run as specific user. Print the first five lines of this directory.

```
ubuntu@ip-172-31-42-193:~$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
```

`tail -n 5 /etc/passwd`: this file is world-readable and lists all system and regular user accounts, including system services that run as specific user. Print the last five lines of this directory.

```
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
tokyo:x:1001:1001::/home/tokyo:/usr/bin/bash
berlin:x:1002:1002::/home/berlin:/usr/bin/bash
professor:x:1003:1003::/home/professor:/usr/bin/bash
nairobi:x:1004:1006::/home/nairobi:/usr/bin/bash
ubuntu@ip-172-31-42-193:~$
```

`view script.sh`: this command is used to view the .sh file in the read-only mode.



```
ubuntu@ip-172-31-42-193: ~
echo "Hello DevOps!"
Hello DevOps!
"script.sh" [readonly] 2L, 22B  2,20  All
```

♦ Task 3: Understand Permissions

- ◆ Format: rwxrwxrwx (owner-group-other)
- ◆ r = read (4), w = write (2), x = execute (1)
- ◆ Check your files: `ls -l devops.txt, notes.txt, scripts.sh`
- ◆ Answer: What are current permissions? Who can read/write/execute?

`ls -l devops.txt, ls -l notes.txt, ls -l script.sh`: list directory contents ‘-l’ flag is used for long listing format.

```
Last login: Mon Feb  9 10:06:09 2026 from 1.38.220.50
ubuntu@ip-172-31-42-193:~$ ls -l devops.txt
-rw-rw-r-- 1 ubuntu ubuntu 0 Feb  9 09:51 devops.txt
ubuntu@ip-172-31-42-193:~$ ls -l notes.txt
-rw-rw-r-- 1 ubuntu ubuntu 36 Feb  9 09:52 notes.txt
ubuntu@ip-172-31-42-193:~$ ls -l script.sh
-rw-rw-r-- 1 ubuntu ubuntu 22 Feb  9 09:53 script.sh
ubuntu@ip-172-31-42-193:~$
```

The current permissions for devops .txt file are owner can read and write, group can also read and write, but others only have read permission.

The current permissions for notes.txt file are owner can read and write, group can also read and write, but others only have read permission.

The current permissions for script.sh file are owner can read and write, group can also read and write, but others only have read permission.

- ◆ Task 4: Modify Permissions
- ◆ Make script.sh executable run it with `./script.sh`
- ◆ Set devops.txt to read only (remove write for all)
- ◆ Set notes.txt to 640 (owner: rw, group: r, others: none)
- ◆ Create directory project with permissions 755
- ◆ Verify `ls -l` after change.

chmod 775 script.sh: This command is used to change file mode bits. Give desired permissions to the owner, group, and others.

./script.sh: This command is used to execute the '.sh' file.

```
ubuntu@ip-172-31-42-193:~$ chmod 775 script.sh
ubuntu@ip-172-31-42-193:~$ ./script.sh
Hello DevOps!
ubuntu@ip-172-31-42-193:~$
```

chmod 444 devops.txt: this command is used to set permissions to read only.

```
ubuntu@ip-172-31-42-193:~$ chmod 444 devops.txt
ubuntu@ip-172-31-42-193:~$ ls -l devops.txt
-r--r--r-- 1 ubuntu ubuntu 0 Feb  9 09:51 devops.txt
ubuntu@ip-172-31-42-193:~$
```

chmod 640 notes.txt: this command is used to set owner permission to read and write, and for group only read permission, no permission for others.

```
ubuntu@ip-172-31-42-193:~$ chmod 640 notes.txt
ubuntu@ip-172-31-42-193:~$ ls -l notes.txt
-rw-r----- 1 ubuntu ubuntu 36 Feb  9 09:52 notes.txt
ubuntu@ip-172-31-42-193:~$
```

mkdir project: this command is used to create a directory with the name project.

chmod 755 project: this command is used to set permissions for owner read, write, execute is given, for group read and execute is given, for others read and execute is given.

```
ubuntu@ip-172-31-42-193:~$ ls -l
total 16
-r--r--r-- 1 ubuntu ubuntu 0 Feb  9 09:51 devops.txt
-rw-rw-r-- 1 ubuntu ubuntu 0 Feb  8 14:58 my_file.txt
-rw-r----- 1 ubuntu ubuntu 36 Feb  9 09:52 notes.txt
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb  9 11:13 project
-rwxrwxr-x 1 ubuntu ubuntu 22 Feb  9 09:53 script.sh
-rw-rw-r-- 1 ubuntu ubuntu 24 Feb  8 15:06 test_file.txt
ubuntu@ip-172-31-42-193:~$
```

◆ Task 5: Test Permissions

- ◆ Try writing a read-only – what happens?
- ◆ Try executing a file without execute permission
- ◆ Document the error messages

echo "hey this is rohan, im learning devops engineering from shubam ." | devops.txt: this command will throw error because this file has only read permission.

```
ubuntu@ip-172-31-42-193:~$ echo "hey this rohan,im learning devops engineeing
with shubham." | devops.txt
devops.txt: command not found
ubuntu@ip-172-31-42-193:~$
```

chmod 644 script.sh: this command is used to give permissions read and write for owner, for group read-only, for others also read-only.

./script.sh: this command is used to execute the '.sh' file.

```
ubuntu@ip-172-31-42-193:~$ chmod 644 script.sh
ubuntu@ip-172-31-42-193:~$ ./script.sh
-bash: ./script.sh: Permission denied
ubuntu@ip-172-31-42-193:~$
```

Documenting the error messages:

chmod 775 script.sh

chmod 775 devops.txt

```
ubuntu@ip-172-31-42-193:~$ chmod 775 script.sh
ubuntu@ip-172-31-42-193:~$ ./script.sh
Hello DevOps!
ubuntu@ip-172-31-42-193:~$
```

```
ubuntu@ip-172-31-42-193:~$ chmod 775 devops.txt
ubuntu@ip-172-31-42-193:~$ ls -l devops.txt
-rwxrwxr-x 1 ubuntu ubuntu 0 Feb  9 09:51 devops.txt
ubuntu@ip-172-31-42-193:~$
```